Koblitz n=83, ℓ=6473

Curve $y^2+xy=x^3+1$ over F_(2^83): t=-6151469093347, D_K=-7, f=6473·53676929, #E=2^2·241785163923079621668568 9. Native koblitz-step built $\Phi_{6473}(X,j_0)$ from 9864 CRT primes (606365-bit coefficients, 8590 s total on 3 threads), reduced mod 2, factored into 78 Frobenius orbits of size 83, and found 6474 distinct level-1 j-invariants in F_(2^83) (expected 6474). 48 of 48 sampled neighbours had exactly one twist of order N; 0 of 8 random j passed the control. ALL CHECKS PASS. j-list and log hashes: SHA256SUMS. Not done: ecbench measurements at n=83 (the group is far outside rho reach) and no isogeny maps; the level-1 neighbours share N, so this verifies the volcano structure at ℓ=6473 only.

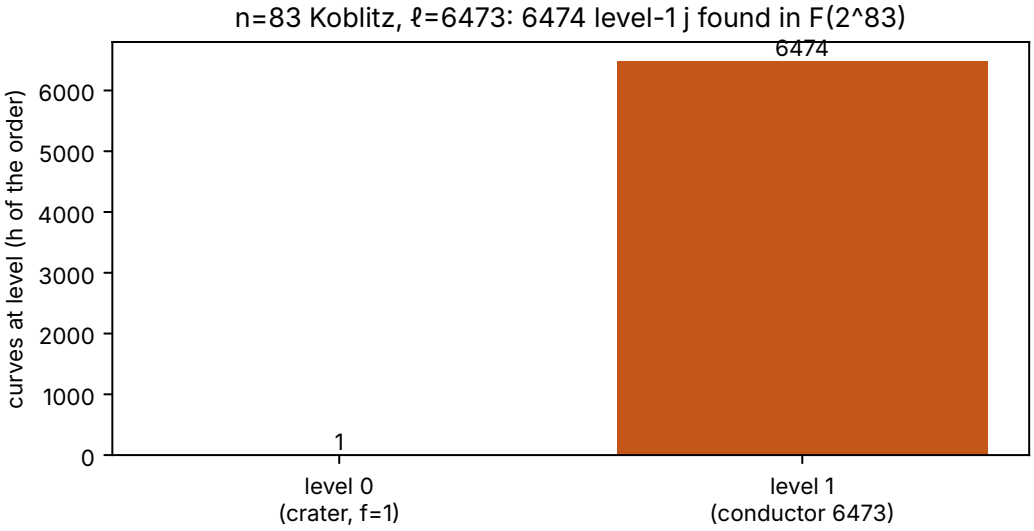

Figure 1: Level sizes found.